Empirical Analysis of ExxonMobil (XOM)
Stylized Facts, Volatility Dynamics and Asset Pricing Models

A quantitative study of nearly 40 years of daily and monthly return data, combining descriptive econometrics, GARCH modelling, and factor-based asset pricing.

Asset	ExxonMobil Corporation (XOM, NYSE)
Sample	December 1985 – December 2024
Frequencies	Daily, Monthly, Annual
Tools	Python · statsmodels · arch · yfinance

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2025

All data sourced from Yahoo Finance and the Ken French Data Library.
Code available at: `https://github.com/enzo-thuilliez/xom-econometric-analysis`

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Overview

This report presents an empirical econometric analysis of ExxonMobil Corporation (XOM), one of the largest publicly traded energy companies in the world. Using nearly four decades of daily adjusted closing prices (December 1985 to December 2024), the study investigates the statistical properties of financial returns and tests two fundamental asset-pricing frameworks.

The analysis is structured in three main parts. The first examines the well-known stylized facts of financial time series — including non-normality, heavy tails, volatility clustering, and leverage effects — through both graphical evidence and formal statistical testing. The second estimates and compares the Capital Asset Pricing Model (CAPM) and the Fama–French Three-Factor Model (FF3) using monthly excess returns. The third, presented as an extension, fits a GARCH(1,1) model to capture time-varying volatility and derives a conditional Value-at-Risk (VaR) estimate.

The entire analysis is implemented in Python and is fully reproducible from the accompanying notebook.

Data Description and Preliminary Steps

The analysis relies on the daily adjusted closing prices of ExxonMobil Corporation (ticker: XOM), traded on the New York Stock Exchange. The sample spans from 31 December 1985 to 31 December 2024, yielding approximately 9,828 trading days — more than 39 years (468 months) of data with no missing trading days. The data were downloaded directly from Yahoo Finance using Python.

Daily logarithmic returns were computed as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right),$$

and then aggregated to monthly and annual frequencies to study the effect of time aggregation on distributional properties.

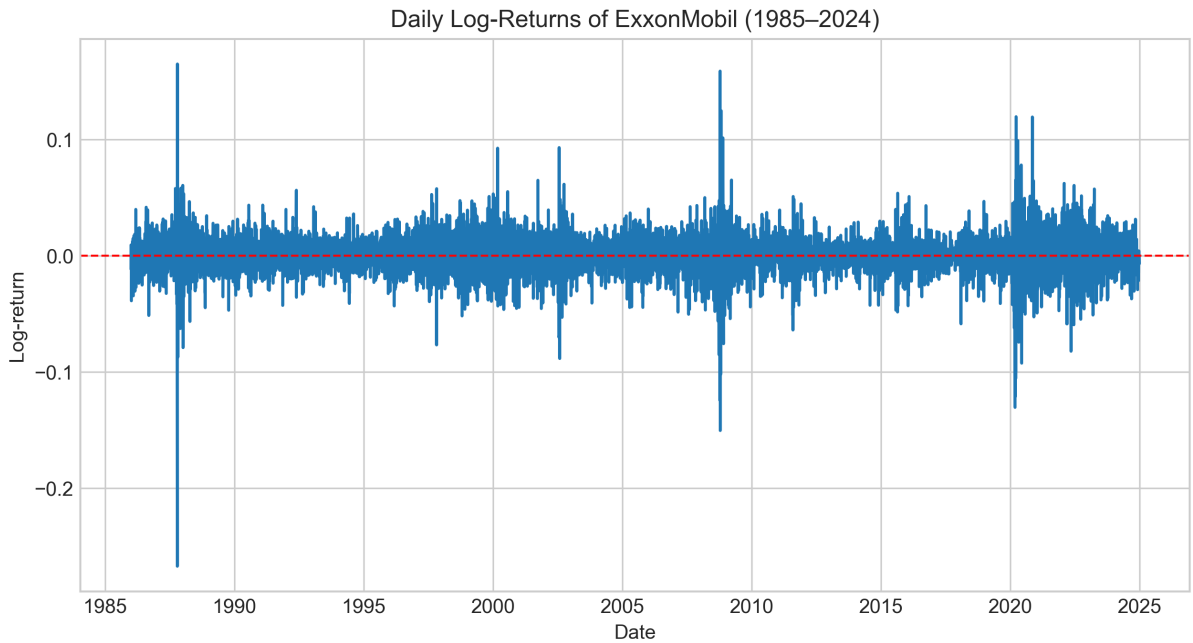

Figure 1: Daily log-returns of ExxonMobil (XOM), 1985–2024.

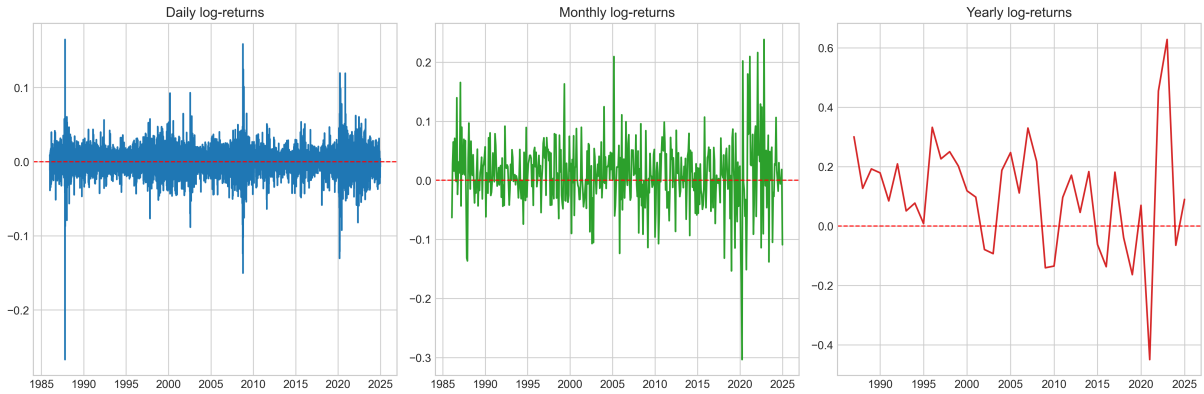

Figure 2: Daily, monthly, and annual log-returns of ExxonMobil (XOM), 1985–2024.

Descriptive Statistics of Returns

The summary statistics in Table 1 provide an overview of the distributional features of ExxonMobil's returns across daily, monthly, and annual frequencies.

At the daily level, the mean return is near zero (0.04% per day), as expected for high-frequency financial data. The standard deviation of 1.6% reflects substantial short-term volatility. Higher-order moments highlight strong deviations from normality: skewness of -0.42 indicates a mild tendency toward larger negative returns, while kurtosis of 16.28 signals extremely fat tails, consistent with frequent extreme price movements.

At the monthly frequency, the mean rises to 0.88% and volatility to 5.7%. Kurtosis drops sharply to 3.50 — much closer to the Gaussian benchmark of 3 — and skewness becomes nearly symmetric (-0.11). At the annual level, mean return reaches 10.5%, volatility stands at 19.1%, and kurtosis falls further to 1.68, indicating a platykurtic distribution. These results confirm *aggregational Gaussianity*: the return distribution converges toward normality as the observation horizon lengthens.

Stylized Facts in Financial Returns

Financial return series share several empirical regularities known as stylized facts. This section examines these properties through graphical evidence and statistical testing.

Weak Stationarity

As shown in Figure 1, daily log-returns fluctuate around a stable mean close to zero throughout 1985–2024. No persistent drift is observed, consistent with weak stationarity. Short-term volatility varies, but the series reverts to its long-term level after episodes of market turbulence — notably around the 2008–2009 global financial crisis and the 2020 COVID-19 shock.

This distinction between non-stationary prices and stationary returns is foundational in financial econometrics, justifying the use of returns — rather than prices — in all subsequent modelling.

Non-Normality and Heavy Tails

Financial return distributions rarely follow the normal law. For ExxonMobil, this is evident in the histograms and Q–Q plots above. Daily returns exhibit a sharply peaked distribution with very heavy tails. Normality tests strongly reject the null for daily (Jarque–Bera = 108,729, $p < 0.001$; Lilliefors $p = 0.001$) and monthly data, but not for annual returns, consistent with aggregational Gaussianity.

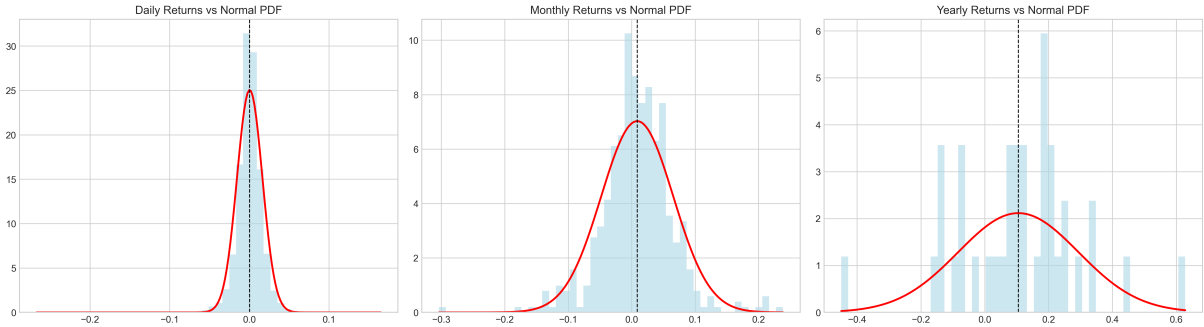

Figure 3: Distribution of daily, monthly, and annual log-returns with fitted normal density (1985–2024).

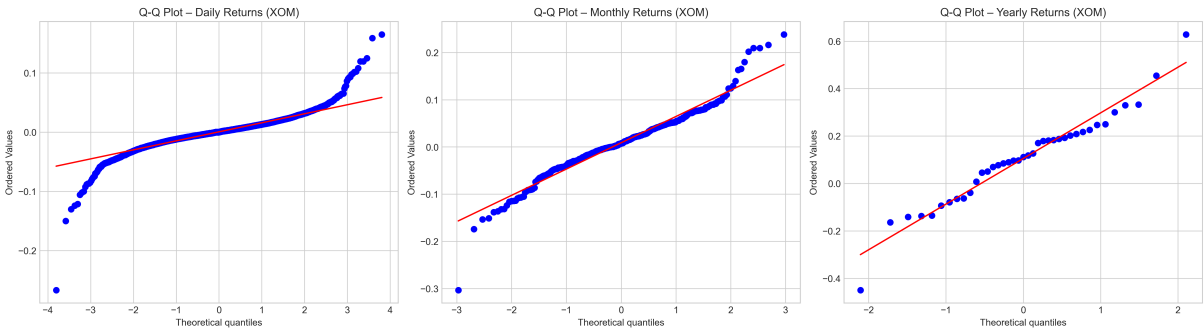

Figure 4: Q–Q plots of daily, monthly, and annual log-returns against the normal distribution.

Time-Varying Moments (Rolling Skewness and Kurtosis)

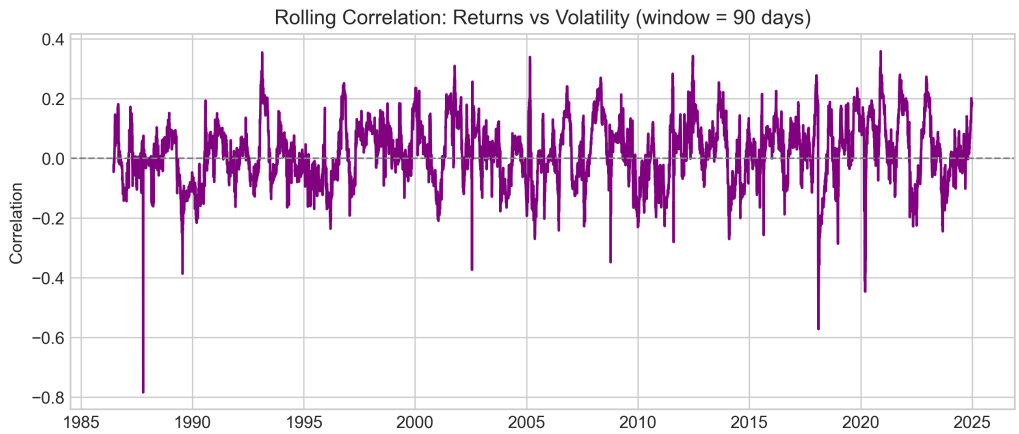

Figure 5: Rolling 252-day skewness and excess kurtosis of daily returns for ExxonMobil (XOM), 1985–2024.

Rolling skewness fluctuates around zero, with periods of strongly negative values during major downturns (2008–2009, 2020), indicating that large negative returns were more frequent during those episodes. Rolling excess kurtosis displays pronounced spikes often between 6 and 10, corresponding to episodes of extreme market stress. Between crises, kurtosis reverts to moderate levels, showing that extreme return behavior is episodic rather than persistent.

Aggregational Gaussianity

As returns are aggregated over longer horizons, their distribution approaches Gaussian. This manifests as: (i) thinner tails and a smoother central peak; (ii) progressive alignment in Q–Q

plots; and (iii) declining kurtosis — from 16.28 (daily) to 3.50 (monthly) to 1.68 (annual). The comparative evidence from histograms, Q–Q plots, and higher moments consistently supports this stylized fact.

Absence of Linear Autocorrelation

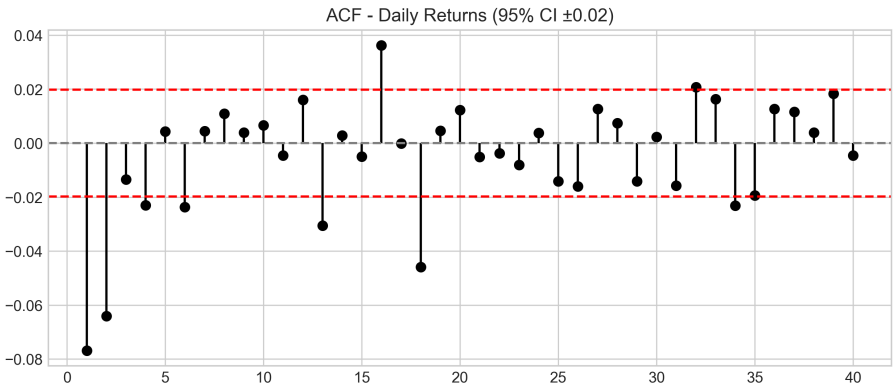

Figure 6: Autocorrelation function (ACF) of daily log-returns of ExxonMobil (XOM).

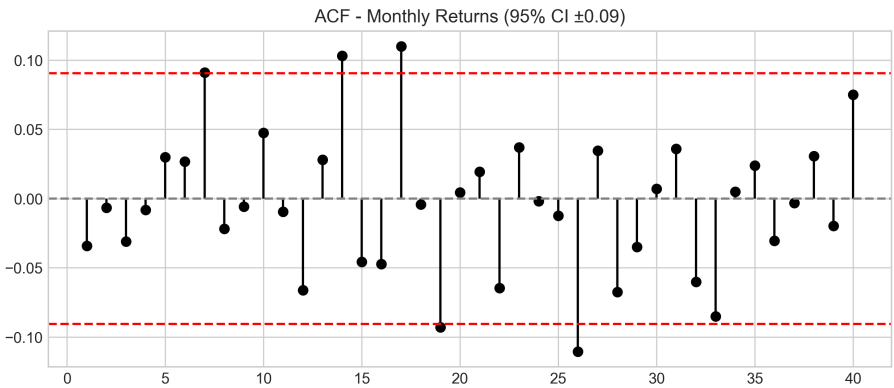

Figure 7: Autocorrelation function (ACF) of monthly log-returns of ExxonMobil (XOM).

Autocorrelation coefficients of raw returns are close to zero at all lags, with most falling within the 95% confidence bounds. This confirms that successive price changes are largely independent, supporting the weak form of market efficiency. The Ljung–Box test yields a near-zero p-value for daily returns, though this largely reflects large-sample sensitivity with nearly 10,000 observations.

Volatility Clustering and Leverage Effect

The autocorrelation of squared and absolute returns remains significantly positive over several lags — a clear signature of conditional heteroskedasticity. The 30-day rolling volatility further illustrates this clustering: periods of high volatility coincide with major market events (2008 financial crisis, 2020 oil price collapse), followed by extended periods of relative calm.

The Ljung–Box test applied to squared and absolute returns rejects the null of no autocorrelation ($p < 0.001$), and the ARCH–LM test strongly confirms conditional heteroskedasticity (statistic $= 1518.3$, $p < 0.001$).

To examine the leverage effect, contemporaneous and lagged correlations between returns and volatility were computed: $\rho = 0.0116$ and $\rho_{\text{lag}} = 0.0039$. Both values are close to zero, suggesting that ExxonMobil's volatility does not respond asymmetrically to negative price shocks. This

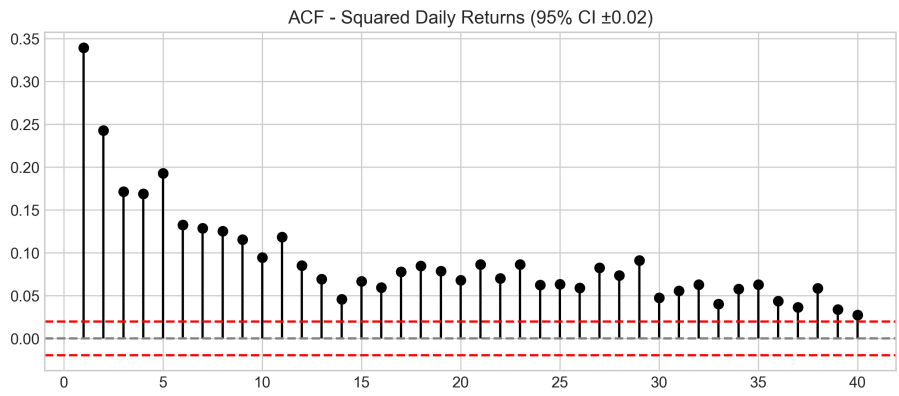

Figure 8: Autocorrelation function (ACF) of squared daily returns of ExxonMobil (XOM).

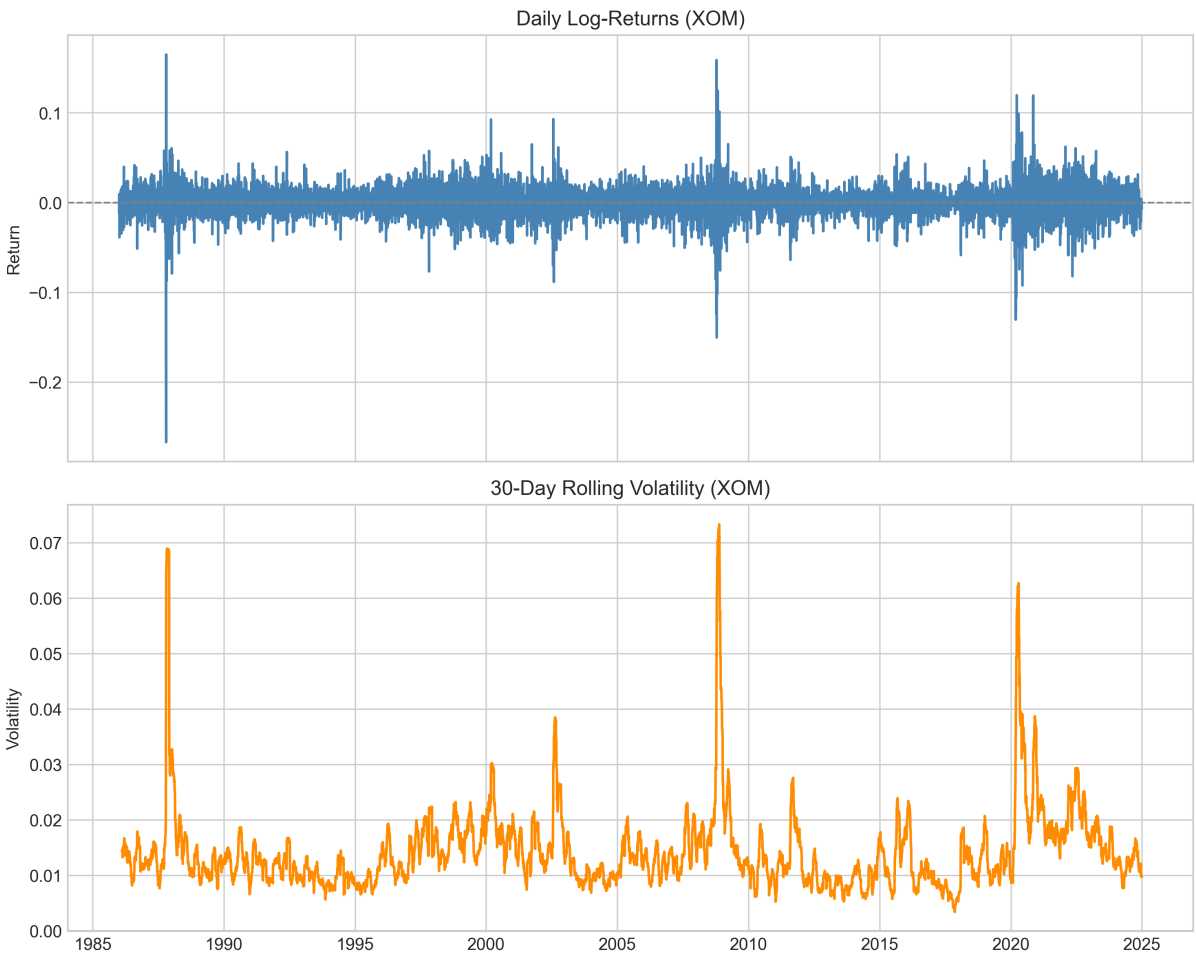

Figure 9: 30-day rolling volatility of ExxonMobil (XOM) daily returns.

absence of a significant leverage effect is consistent with observations for individual stocks, where firm-specific dynamics often dominate financial leverage asymmetries.

Asset Pricing Models: CAPM vs. Fama–French Three Factors

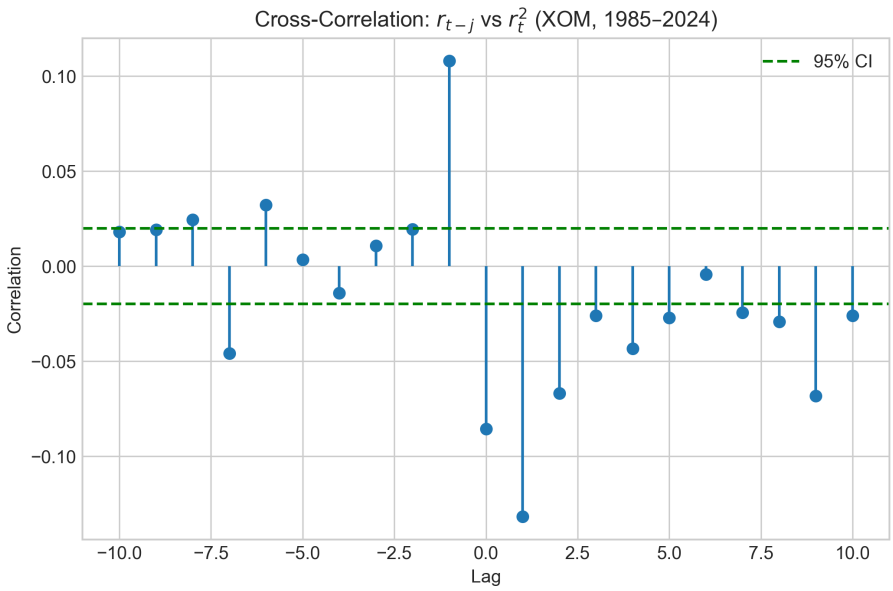

Figure 10: Cross-correlation between lagged returns (r_{t-j}) and current squared returns (r_t^2).

Data and Setup

Monthly excess returns of ExxonMobil are aligned with the Fama–French research factors Mkt–RF, SMB, and HML. The final sample spans January 1986 to December 2024 ($N = 468$ months). Prices are sampled at the last trading day of each month and converted to simple returns. The Fama–French factors, including the risk-free rate RF, are expressed in percentage points per month. XOM's excess return is computed as:

$$R_{i,t} - R_{f,t} = 100 \times \left(\frac{P_t}{P_{t-1}} - 1 \right) - \mathrm{RF}_t.$$

Estimation is by ordinary least squares with an intercept. The merged dataset contains no missing values and has final shape $(468, 7)$.

Model Specifications

CAPM (market model):

$$R_{i,t} - R_{f,t} = \alpha + \beta_{\mathrm{MKT}}(R_{M,t} - R_{f,t}) + u_t.$$

Fama–French Three-Factor Model:

$$R_{i,t} - R_{f,t} = \alpha + \beta_{\mathrm{MKT}}\,\mathrm{Mkt\text{-}RF}_t + \beta_{\mathrm{SMB}}\,\mathrm{SMB}_t + \beta_{\mathrm{HML}}\,\mathrm{HML}_t + u_t.$$

Estimation Results

Interpretation and Comparison

Market exposure is economically meaningful and highly significant in both specifications. In the CAPM, $\hat{\beta}_{\mathrm{MKT}} \approx 0.629$ implies that a 1 percentage point increase in the market excess return is associated on average with 0.63 pp in XOM's excess return. When SMB and HML are added, the market loading rises to 0.703, as part of the comovement initially attributed to the market is reallocated to the size and value dimensions.

The value loading is large and precisely estimated ($\hat{\beta}_{\mathrm{HML}} = 0.531$, $p < 0.001$), consistent with the characteristics of a mature, capital-intensive energy firm. The size loading is modest and

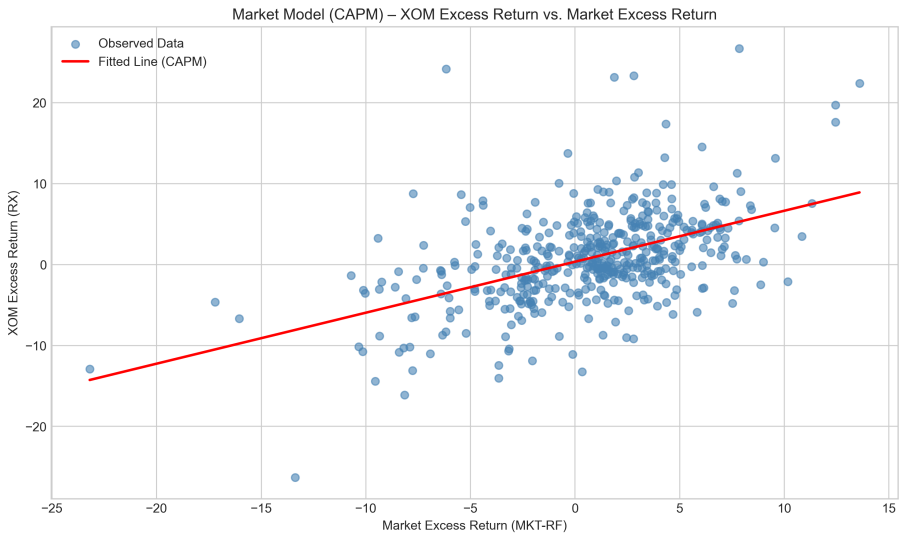

Figure 11: Market model (CAPM): XOM excess return (%) vs. Mkt–RF (%). OLS fitted line in red.

only marginally significant ($\hat{\beta}_{\text{SMB}} = -0.130$, $p \approx 0.076$), in line with ExxonMobil's large-cap profile.

In both models, the intercept is statistically indistinguishable from zero ($p \approx 0.17$ in CAPM, $p = 0.368$ in FF3), indicating no evidence of abnormal mean excess return once systematic risk exposures are accounted for.

Goodness of fit improves from the single-index model to the three-factor model: R^2 increases from 0.246 to 0.339. The gain is primarily driven by the HML factor, while SMB adds limited incremental explanatory power in this sample.

Extension: Conditional Volatility and Value-at-Risk (GARCH)

GARCH(1,1) Model

The strong evidence of volatility clustering and conditional heteroskedasticity identified above motivates a parametric volatility model. A GARCH(1,1) is estimated on daily returns (in percentage form):

$$r_t = \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_t^2)$$
$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2$$

where α measures the short-term impact of shocks (ARCH effect), β captures volatility persistence (GARCH effect), and ω represents the long-run variance level.

The estimated coefficients satisfy the stationarity condition $\alpha + \beta < 1$, with $\alpha + \beta = 0.988$. This high persistence implies that shocks to volatility dissipate slowly — consistent with the slow decay observed in the ACF of squared returns.

The conditional volatility series captures well-known episodes of market stress: the 1987 Black Monday crash, the 2008 global financial crisis, and the 2020 COVID-19 oil price collapse. The GARCH-based estimate is smoother and more persistent than a simple rolling standard deviation.

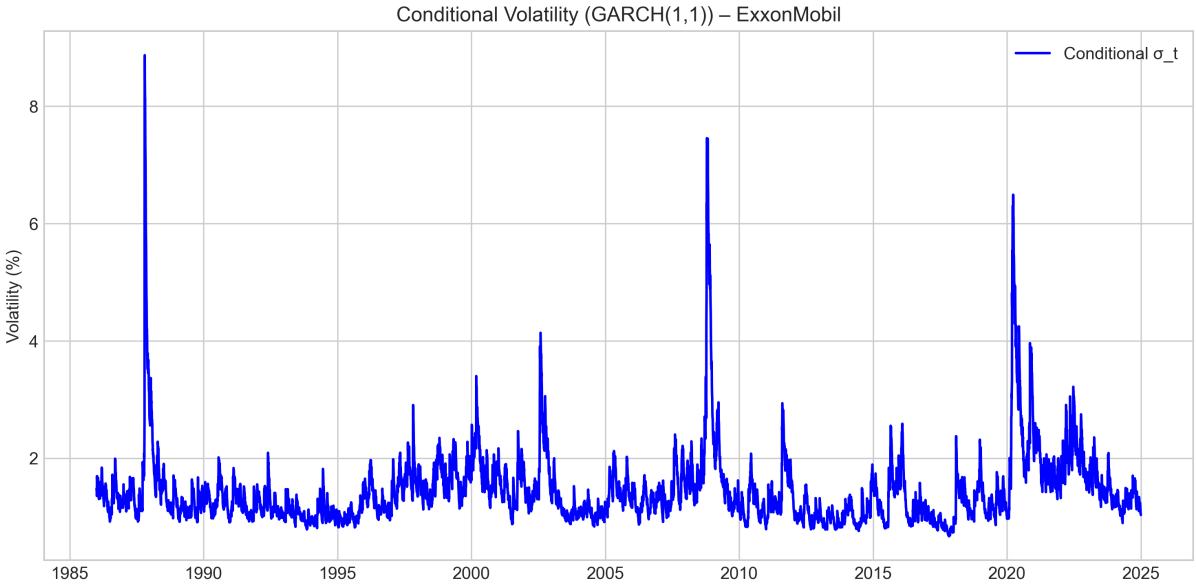

Figure 12: Conditional volatility estimated by the GARCH(1,1) model for ExxonMobil (XOM), 1985–2024.

Conditional Value-at-Risk

A 1-day-ahead VaR at the 95% confidence level is computed as:

$$\mathrm{VaR}_{t,\,95\%} = \sigma_t\, \Phi^{-1}(0.05)$$

where $\Phi^{-1}(0.05)$ is the 5th percentile of the standard normal distribution. The resulting estimate is approximately -1.67%.

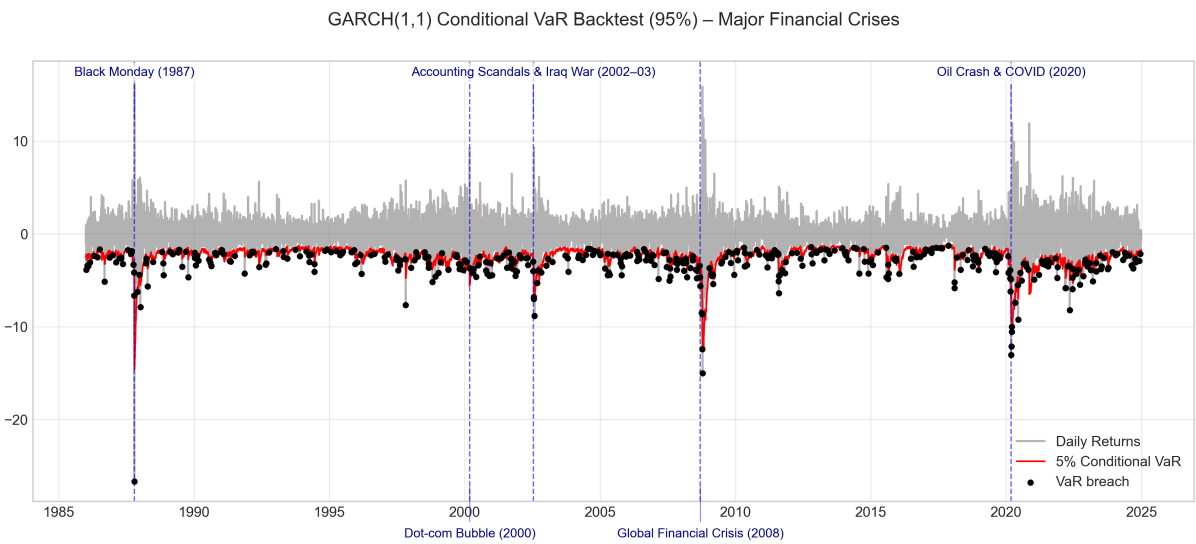

Figure 13: Backtest of the 5% conditional VaR (GARCH(1,1)) with observed daily returns, highlighting major financial crises.

Backtesting confirms that approximately 5% of observations fall below the estimated VaR threshold (empirical coverage = 4.57%), consistent with the theoretical expectation. Periods with numerous violations correspond to global crises, highlighting the model's ability to adjust volatility forecasts dynamically. A GARCH(1,1) with Student-t innovations could further improve calibration during tail events.

Conclusion

This analysis provides a comprehensive empirical characterization of ExxonMobil's return dynamics over nearly four decades. The key findings are as follows.

In terms of stylized facts, daily returns exhibit strong non-normality (kurtosis = 16.28), pronounced volatility clustering, and near-zero linear autocorrelation. These features are consistent with the canonical properties documented in the financial econometrics literature. The leverage effect is absent, as expected for a single large-cap stock where firm-specific dynamics dominate.

In terms of asset pricing, both the CAPM and the Fama–French three-factor model confirm a meaningful market beta around 0.63–0.70 and a pronounced value tilt ($\hat{\beta}_{\text{HML}} \approx 0.53$), consistent with ExxonMobil's profile as a mature, asset-intensive energy firm. No statistically significant alpha is detected in either specification. The FF3 model achieves a higher R^2 (0.34 vs. 0.25), primarily due to the HML factor.

Finally, the GARCH(1,1) extension confirms time-varying volatility with high persistence ($\alpha + \beta \approx 0.988$) and produces reliable conditional VaR estimates, as verified by backtesting.

	Mean	Std Dev	Skewness	Kurtosis	Min	5%	25%	Median	75%	95%	Max	JB stat	JB p-val	Lillie stat	Lillie p-val	N obs
Daily	0.0004	0.0159	-0.4183	16.2829	-0.2669	-0.0232	-0.0077	0.0000	0.0086	0.0237	0.1648	108729.2647	0.0000	0.0627	0.0010	9827.0000
Monthly	0.0088	0.0568	-0.1100	3.5047	-0.3036	-0.0898	-0.0203	0.0076	0.0421	0.0887	0.2384	233.6463	0.0000	0.0603	0.0010	468.0000
Annual	0.1053	0.1911	-0.1246	1.6775	-0.4496	-0.1428	-0.0152	0.1114	0.2057	0.3446	0.6281	2.9246	0.2317	0.0959	0.4825	39.0000

Table 1: Ljung–Box and ARCH–LM tests for daily returns of ExxonMobil (XOM).

Test	Statistic	p-value
Ljung–Box (r_t, 20 lags)	—	0.000
Ljung–Box ($\lvert r_t \rvert$, 20 lags)	—	0.000
Ljung–Box (r_t^2, 20 lags)	—	0.000
ARCH–LM (10 lags)	1518.3	< 0.001

Table 2: CAPM on XOM monthly excess returns — OLS.

	Coefficient	t-stat	p-value
Intercept (α)	0.329	1.40	0.160
β_{MKT}	0.629	12.32	0.000
N		468	
R^2		0.246	
Adj. R^2		0.244	

Table 3: Fama–French Three-Factor Model on XOM monthly excess returns — OLS.

	Coefficient	t-stat	p-value
Intercept (α)	0.198	0.900	0.368
β_{MKT}	0.703	14.206	0.000
β_{SMB}	-0.130	-1.780	0.076
β_{HML}	0.531	7.551	0.000
N		468	
R^2		0.339	
Adj. R^2		0.335	